

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)**(TRIMESTER –IV) END TERM EXAMINATION, OCTOBER-2023****Inventory and Warehouse Management****[Time: 2 Hours 30 Minutes]****[Maximum Marks: 70]**[**BL** – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)
CO – Course Outcomes]**Following are the course outcomes of the subject: - Inventory and Warehouse Management**

CO	Course Outcome (CO)
CO1	To understand the basic concept and various function of warehouse
CO2	To impart knowledge on various types of warehouses and their advantages.
CO3	To study the identification of various material and material storage systems, material handling equipment

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)**

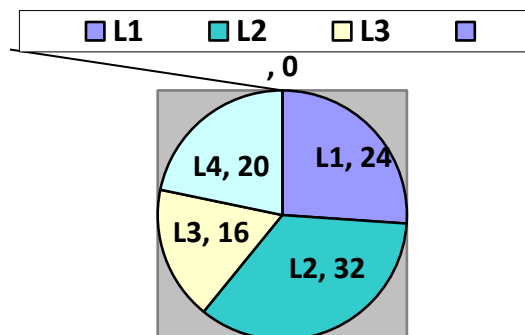
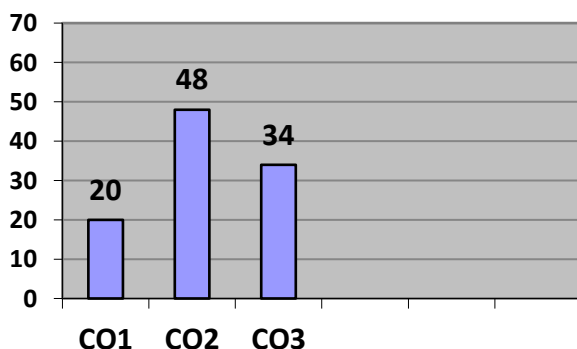
Q.1	Marks	CO	BL
(A)	4	CO - 1	L - 4
(B)	4	CO - 1	L - 1
(C)	4	CO - 1	L - 3
(D)	4	CO - 1	L - 1
(E)	4	CO - 1	L - 3

SECTION-B**Attempt all questions out of the following: (Long Answer Type Questions)**

Q. No	Marks	CO	BL
2 (A)	8	CO - 2	L - 2
2 (A)	8	CO - 2	L - 1
2 (B)	8	CO - 2	L - 3
2 (A)	8	CO - 2	L - 1
3 (A)	8	CO - 2	L - 4
3 (A)	8	CO - 2	L - 2
3 (B)	8	CO - 3	L - 2
3 (B)	8	CO - 3	L - 2

SECTION-C**Attempt all parts of the following: (Caselet / Numerical)**

Q.4	Marks	CO	BL
4 (A)	9	CO - 3	L - 4
4 (B)	9	CO - 3	L - 4

Course Outcome Wise Marks Distribution**Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2022-24)

(TRIMESTER – IV) END TERM EXAMINATION, OCTOBER - 2023

Inventory and Warehouse Management

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 70]

INSTRUCTIONS:

- (i) **There are THREE sections in this question paper.**
- (ii) **Attempt questions from each section as per the directions.**
- (iii) **Make suitable assumptions wherever necessary.**

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) The integration of AI & Automation in warehouse industry is changing the game. What changes in warehouse industry do you see. Support your answer with a case.
- (b) Warehousing is a complex activity. Its basic components shall be handled effectively and efficiently. Do you agree? Give your answer along the explanation of all the components.
- (c) You are hired as a supply chain manager in a start-up that manufactures shoes. The company already have 2 warehouses but the operations at the warehouses are not running effectively. Your task is to develop the right strategy for the warehouses. How would you complete the task?
- (d) Identify the following racking systems and explain their workings.

i



ii



- (e) As a supply chain manager you have given the responsibility to identify a new location for setting up a warehouse. What factors would you consider when choosing the warehouse?

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(4×8=32)

- (a) Inventory is one of the biggest assets of any business, and a warehouse is a place where inventories are managed & stored. Considering the above principles, an efficient layout of the facility can help the firms to save costs, create a more efficient workplace, make the most of the available space, and most importantly create a better experience for workers & customers. Do you agree with the statement. Support your answer by explaining the factors that impact warehouse design.

OR

- (a) All warehouses should have these areas in order to run effectively. What are these areas?
- (b) In Order to operate a warehouse, safety plays a crucial role. Explain the guidelines to keep the warehouse safe.

OR

- (b) Warehouse provide protection to goods against heat, wind, storm, moisture, etc. and also cut down losses due to spoilage, wastage etc. This is the basic function of every warehouse. In addition to this, warehouses nowadays also perform a variety of other functions.. Shed light on all the functions performed in a warehouse.

Q.3. Attempt all questions of the following

- (a) Advancement in technology has allowed for the putaway process to move goods for storage to their most optimal location in a fast, efficient, and effective manner. Do you agree? If yes, explain different types of putaway.

OR

- (a) There are many ways in which organizations prefer to manage their picking process. Explain the commonly adopted picking methods.
- (b) In a warehouse environment, reverse logistics pertain to any of the following activities after the initial purchase. Identify and explain these activities, also write about the KPI's of warehouse operations.

OR

- (b) There are different types of material handling equipment that directly impact the operations in a warehouse. There are 4 major categories. Identify and explain these categories. Also explain atleast 2 MHE under each category.

SECTION-C

Q.4. Attempt all parts of the following:

(2×9=18)

Optimizing Warehouse Management for Gagan Logistics

Background:

Gagan Logistics is a leading third-party logistics provider specializing in the storage and distribution of goods for various clients across multiple industries. With a growing client base and increasing inventory demands, Gagan Logistics is facing challenges in its warehouse management system. The company is tasked with finding the most efficient way to handle its diverse inventory, which ranges from electronics and perishables to heavy machinery.

Objective:

Gagan Logistics is seeking a comprehensive solution to improve its warehouse management system, optimize inventory storage, streamline order fulfillment, and enhance overall operational efficiency. This case study explores various aspects of warehouse management, including centralized and decentralized warehousing, warehouse management systems (WMS), storage policies (class-based and dedicated), and picking and putaway systems.

Gagan Logistics needs to evaluate whether to continue its centralized warehousing approach or adopt a decentralized model. The centralized warehouse is located in a single, large facility, while the decentralized approach involves multiple smaller warehouses located closer to the end customers. The company must decide on the implementation of a WMS to automate and optimize various warehouse operations. Key considerations include the selection of the appropriate WMS software, integration with existing systems, and the training of staff. Gagan Logistics must develop a storage policy that suits its diverse range of products.

Challenges:

Gagan Logistics faces several challenges in optimizing its warehouse management system:

1. **Inventory Diversification:** The company handles a wide range of products with varying storage and handling requirements.
2. **Order Fulfillment Efficiency:** Streamlining order picking and packing processes to reduce lead times and errors is critical.
3. **Operational Scalability:** The solution should allow for future growth and adaptability to changing client needs.
4. **Cost Management:** Balancing cost-effectiveness with service quality is essential.

Conclusion:

Gagan Logistics faces critical decisions regarding its warehouse management system, including warehouse structure, WMS implementation, storage policies, and picking/putaway strategies. By carefully examining these aspects and considering the unique characteristics of their inventory and operations, Gagan Logistics can optimize its warehouse management system and continue to provide high-quality services to its clients while ensuring cost-effectiveness and scalability.

- 4(a)** Compare and contrast the advantages and disadvantages of centralized and decentralized warehousing for Gagan Logistics. What factors should influence their decision? Also Discuss the benefits of implementing a Warehouse Management System (WMS) for Gagan Logistics. What features and functionalities should they look for in a suitable WMS?
- 4(b)** Analyze the suitability of class-based and dedicated storage policies for Gagan Logistics. Which policy would be more effective for their diverse inventory? Propose a comprehensive warehouse management solution that addresses Gagan Logistics' challenges and objectives, taking into account inventory diversification, order fulfillment efficiency, operational scalability, and cost management.